CONTACT: Brian Gorberg

This project was created to help organize and streamline the flood modeling process. It is used to view USGS stream gauge data, calculate probabilities, and determine rating curves. This is then often applied to flood models for calibration, validation, and scenario testing.

In this case, we use three examples of USGS gauges in Hawaii (Waiahole, Waialua, Hanalei). We use 4 scripts to run through a modeling analysis. The scripts streamline the process of gathering metadata, pulling observed annual peaks, analyzing the probability, calculating rating curves, and then estimating model uncertainty by comparing observations to model outputs.

YOU MUST DOWNLOAD PEAK FQ OUTPUTS FOR THIS ALL USGS GAUGES METADATA FOR HAWAII IS INCLUDED IN THE FOLDER PULL_USGS_METADATA**

YOU MUST FOLLOW THE RATING CURVE GITHUB REPOSITORY TO CREATE A KERNEL IN JUPOYTER NOTEBOOKS

!!!YOU SHOULD MAKE A NEW ENVIRONMENT!!!

SEE SOURCE:

https://github.com/thodson-usgs/ratingcurve

The scripts do the following:

1) Select USGS gauges to get metadata and read peakfq results
2) Derive rating curves for the annual peaks from the peak fq results ^
3) Observed rating curves are cleaned into incremental stage values (1ft), the discharge is calculated from the rating curves, and then probabilities are estimated using the Peak Fq probability output
4) An example is provided to compare the modeled and observed rating curves. Uncertainty in the model is calculated using the observed rating curves and the GSSHA flood model's stage/discharge relationship.

Functions are found in the .py files

Model_results contain the GSSHA files for (4)

SAVE_DATA contains all script outputs

PULL_USGS_METADATA contains a script to retrieve Hawaii gauges, but also a file of retrieved hawaii gauges

SAVE DATA:

USGS_Gauge_Metadata = (1)

Peak_Fq_USGS_probability = (1)

Stream_Gauge_Annual_Peaks-input = (1)

OBSERVED-rating-curves-results = (2)

OBSERVED-stage-discharge-probability-results = (3)

GSSHA-MODEL-observed-stage-uncertainty = (4)

GSSHA-MODEL_stage-discharge-probability-results = (4)

THE REST OF THIS DOCUMENT… contains AI generated text for more information on all files and figures within this github repository.

Rating Curve Probability

Technical README & reproducibility guide

Repository inventory • notebook-to-output lineage • figures • table schemas • calculation definitions

Source: Rating_Curve_Probability.zip supplied October 2026. This guide documents the files and code present in that archive, not independently verified model correctness.

1. Project overview

The repository combines USGS gauge metadata and annual peak flows, PeakFQ flood-frequency estimates, observed stage–discharge rating-curve fits, annual exceedance probabilities (AEP), and Waialua GSSHA model water-surface elevations. The three study locations are Hanalei (16103000; the Hanalei plot also references 16104200), Waiahole (16294100), and Waialua/Kaukonahua (16210500).

2. Run order and script ownership

Order	Notebook	Role
0	PULL_USGS_METADATA/Get_stream_metadata.ipynb	Download/assemble Hawai'i gauge inventory; prepare peak-record metadata GIS/CSV inputs.
1	1_USGS-meta-peakFq-results.ipynb	Select 3 gauges; read Peak_Fq/*/*_qnt.csv and NWIS peaks; save metadata, flood-frequency table, annual peak CSVs.
2	2_OBSERVED_USGS-rating-curves.ipynb	Fit observed rating curves, compare Waialua old/new datum, save observed curves and diagnostic figures.
3	3_OBSERVED_probability-stage-discharg.ipynb	Merge observed rating curves with PeakFQ AEP estimates; save 3 watershed tables, summary and 2 comparison figures.
4	4_GSSHA-MODEL_compared_w_OBSERVED.ipynb	Read GSSHA Model results, PeakFQ AEP estimates, plot model-vs-observed curve, calculate discharge-matched stage errors.

Reusable Python modules

Module	Key functions / responsibility
USGS_Functions.py	save_gauge_metadata, read_nwis_peak, get_usgs_continuous, get_start_date_instantaneous; NWIS parsing and retrieval.
rating_curve_DISCHARGE_STAGE_PROBABILITY.py	define_observed_rating_curve, plot_rating_curve, load_watershed_data, merge_rating_probability, predict_exceedance_probability, discharge_at_return_period, create_stage_probability_table.
Compare_MODEL_vs_OBSERVED_rating_probability.py	get_stage_flows, get_gssha_rating_curve, predict_gssha_probability, plot_rating_curve_with_gssha, calculate_gssha_stage_error; GSSHA and observed-model comparisons.

3. Key paths and configuration

All notebooks use Path.cwd() as script_dir. Run Jupyter from the repository root so relative folders resolve correctly. Inputs and outputs are organized beneath SAVE_DATA/, Peak_Fq/, model_results/, figures/, and PULL_USGS_METADATA/.

Configuration	Value found in archive	Meaning
GSSHA .ows source	model_results/Waialua/*.ows	One file per modeled discharge; filename embeds flow in ft³/s.
GSSHA steady-state average	tail_records=120	Mean of final 120 WSE records (records, not necessarily minutes).
Waialua elevation offset	offset_m=4.216908	Added to mean GSSHA WSE in meters before converting to stage in feet.
Stage conversion	1 m = 3.280839895 ft	Adjusted GSSHA height -> stage_ft.
Discharge conversion	1 ft³/s = 0.028316846592 m³/s	Used in model and probability tables.
GSSHA probability source	SAVE_DATA/OBSERVED-stage-discharge-probability-results	Input CSV selected by watershed name.
Stage error reference	datum_merged_waialua['rating_curve_model_stage_results']	Observed fitted rating curve inverted at each model flow.
Uncertainty settings	3 segments; 95%; NUTS	Specified in observed rating fitting cells; stage-error output itself has no 95% interval.

4. Figure register — all 11 saved figures

Figure (figures/)	Producer	Interpretation
Curves_datum_differences.png	2_OBSERVED_USGS-rating-curves.ipynb · cell 13	Comparison of old/new Waialua posterior median curves and estimated datum shift.
Gauge_datum_differences.png	2_OBSERVED_USGS-rating-curves.ipynb · cell 14	Stage difference vs discharge for datum adjustment.
Hanalei_curve.png	2_OBSERVED_USGS-rating-curves.ipynb · cell 7	Hanalei observed stage–discharge fit, observations and uncertainty.
merged_discharge_probability_curves.png	3_OBSERVED_probability-stage-discharg.ipynb · cells 3–4	Three-watershed discharge versus annual exceedance probability; PeakFQ points and merged rating values.
merged_stage_discharge_rating_curves.png	3_OBSERVED_probability-stage-discharg.ipynb · cells 3–4	Three-watershed observed rating curves, merged 1-ft stage points and uncertainty.
Waiahole_curve.png	2_OBSERVED_USGS-rating-curves.ipynb · cell 5	Waiahole observed stage–discharge fit.
Waialua_Model_Curve.png	No direct saving call found in active notebooks	Existing archived figure; provenance not established by inspected current source.
Waialua_newer_datum_curve.png	2_OBSERVED_USGS-rating-curves.ipynb · cell 10	Waialua newer-datum rating fit.
Waialua_older_datum_rating.png	2_OBSERVED_USGS-rating-curves.ipynb · cell 11	Waialua older-datum rating fit.
Waialua_rating_curve_adjusted_datums.png	2_OBSERVED_USGS-rating-curves.ipynb · cell 18	Final Waialua merged datum-adjusted observed rating fit.
Waialua_rating_curve_comparative_GSSHA.png	4_GSSHA-MODEL_compared_w_OBSERVED.ipynb · cell 12 (also 4_READ...)	GSSHA model points overlaid on observed fitted rating curve and uncertainty.

5. Saved data tables — complete SAVE_DATA CSV register

SAVE_DATA/GSSHA-MODEL-observed-stage-uncertainty

Source/owner: 4_GSSHA-MODEL_compared_w_OBSERVED.ipynb · calculate_gssha_stage_error

File	Rows	Columns
Waialua_GSSHA_stage_error.csv	14	flow_cfs, flow_cms, GSSHA_height_m, offset_m, adjusted_height_m, stage_ft, incremental_stage_ft, EXC_Prob, return_period_years, observed_stage_ft,

		stage_error_ft, stage_error_m, absolute_stage_error_ft

SAVE_DATA/GSSHA-MODEL_stage-discharge-probability-results

Source/owner: 4_GSSHA-MODEL_compared_w_OBSERVED.ipynb · predict_gssha_probability

File	Rows	Columns
Waialua_GSSHA_FIM_results.csv	14	flow_cfs, flow_cms, GSSHA_height_m, offset_m, adjusted_height_m, stage_ft, incremental_stage_ft, EXC_Prob, return_period_years

SAVE_DATA/OBSERVED-rating-curves-results

Source/owner: 2_OBSERVED_USGS-rating-curves.ipynb · define_observed_rating_curve

File	Rows	Columns
Hanalei_RECORDED_discharge-stage.csv	40	stage, discharge
Hanalei_observed_rating_curve_model.csv	1000	stage_ft, lower_cfs, median_cfs, upper_cfs
Waiahole_RECORDED_discharge-stage.csv	22	stage, discharge
Waiahole_observed_rating_curve_model.csv	1000	stage_ft, lower_cfs, median_cfs, upper_cfs
waialua-datum-adjusted_RECOREDED_discharge-stage.csv	48	stage, discharge
waialua-datum-adjusted_observed_rating_curve_model.csv	1000	stage_ft, lower_cfs, median_cfs, upper_cfs

SAVE_DATA/OBSERVED-stage-discharge-probability-results

Source/owner: 3_OBSERVED_probability-stage-discharg.ipynb · create_stage_probability_table

File	Rows	Columns
Hanalei_stage_discharge_probability.csv	19	stage_ft, stage_m, median_cfs, median_cms, lower_cfs, upper_cfs, EXC_Prob, return_period_years
Waiahole_stage_discharge_probability.csv	18	stage_ft, stage_m, median_cfs, median_cms, lower_cfs, upper_cfs, EXC_Prob, return_period_years
Waialua_stage_discharge_probability.csv	24	stage_ft, stage_m, median_cfs, median_cms, lower_cfs, upper_cfs, EXC_Prob, return_period_years
watershed_probability_summary.csv	3	watershed, target_return_period, target_discharge_cfs, target_stage_ft, cutoff_stage_ft, extra_stage_ft, reached_target

SAVE_DATA

Source/owner: 1_USGS-meta-peakFq-results.ipynb

File	Rows	Columns
Peak_Fq_USGS_probability.csv	25	EXC_Prob, Hanalei_weighted, Waiahole_weighted, Waialua_weighted
USGS_Gauge_Metadata.csv	3	OBJECTID, Latitude, Longitude, STA_ID, STA_NAME, STA_TYPE, COORDSRC, SITESTATUS, REACH100, MEASURE100, REACH24, MEASURE24, REACHLOC, MEASURELOC, FeatureURL, Peak_Start, Peak_End, Peak_Years

SAVE_DATA/Stream_Gauge_Annual_Peaks-input

Source/owner: Pre-existing input CSVs (used by step 2; not generated by its cells)

File	Rows	Columns
Hanalei_Peaks_63-21_upper-lower-gauges_discharge-height.csv	40	Date, Discharge (cfs), Gauge Height (ft)
Hanalei_weighted_peaks.csv	60	agency_cd, site_no, peak_dt, peak_tm, peak_va, gage_ht, water_year
Waiahole_weighted_peaks.csv	22	agency_cd, site_no, peak_dt, peak_tm, peak_va, gage_ht, water_year
Waialua_Peaks_05-23.csv	17	agency_cd, site_no, peak_dt, peak_tm, peak_va, peak_cd, gage_ht, gage_ht_cd, year_last_pk, ag_dt, ag_tm, ag_gage_ht, ag_gage_ht_cd
Waialua_Peaks_63-03.csv	38	Date, Peak Discharge (cfs), Gage Height (ft)
Waialua_weighted_peaks.csv	57	agency_cd, site_no, peak_dt, peak_tm, peak_va, gage_ht, water_year

SAVE_DATA/Stream_Gauge_Annual_Peaks

Source/owner: 1_USGS-meta-peakFq-results.ipynb · read_nwis_peak

File	Rows	Columns
Hanalei_weighted_peaks.csv	60	agency_cd, site_no, peak_dt, peak_tm, peak_va, gage_ht, water_year
Waiahole_weighted_peaks.csv	22	agency_cd, site_no, peak_dt, peak_tm, peak_va, gage_ht, water_year
Waialua_weighted_peaks.csv	57	agency_cd, site_no, peak_dt, peak_tm, peak_va, gage_ht, water_year

6. Column dictionary and interpretation

Field(s)	Meaning
stage_ft / stage_m	Gauge height or rating stage in feet / meters; stage_m = stage_ft × 0.3048.
flow_cfs / flow_cms	GSSHA simulated discharge in ft^3/s / m^3/s. Note: cfs and ft^3/s are equivalent.
median_cfs / median_cms	Median modeled rating-curve discharge at a given stage, in ft^3/s / m^3/s.
lower_cfs / upper_cfs	Lower and upper modeled discharge uncertainty bounds at each stage (typically 95% for the fitted curves). These are discharge bounds, not stage-error bounds.
discharge / discharge_cfs	Recorded flow in ft^3/s; discharge_cfs is also used for the fitted 1-ft stage results.
EXC_Prob	Annual exceedance probability (fraction 0–1); e.g. 0.01 = 1% AEP.
return_period_years	Reciprocal of AEP: T = 1 / EXC_Prob, in years. A statistical recurrence metric, not a deterministic schedule.
GSSHA_height_m	Mean GSSHA WSE over the last tail_records entries in the .ows file, in meters, before offset.
offset_m	Constant offset applied to the GSSHA mean WSE (4.216908 m in Waialua step 4).
adjusted_height_m	GSSHA_height_m + offset_m.
incremental_stage_ft	Sequential stage-label column beginning at the next whole increment; not the actual GSSHA stage and not interpolated.
observed_stage_ft	Observed rating-curve stage estimated by inverse PCHIP at the same GSSHA discharge.

stage_error_ft / stage_error_m	Signed model-reference difference: GSSHA stage_ft − observed_stage_ft; positive = GSSHA higher.
absolute_stage_error_ft	Absolute value of signed stage error; magnitude only.
target_return_period	Requested target return period in years (1000 in the archived watershed summary).
target_discharge_cfs	Discharge corresponding to the requested target return period.
target_stage_ft	First stage grid point meeting target threshold in the merged output.
cutoff_stage_ft / extra_stage_ft	Saved stopping stage and extension beyond the target stage, in feet.
reached_target	Boolean indicating whether the target was reached.
peak_va / gage_ht	USGS annual peak discharge (ft^3/s) and associated gage height (ft).
peak_dt / peak_tm / water_year	Date/time of peak and hydrologic water year.
Peak_Start / Peak_End / Peak_Years	First/last peak dates and length of record metadata.
STA_ID / STA_NAME / Latitude / Longitude	USGS site identifier, station name, and station coordinates.
OBJECTID / STA_TYPE / COORDSRC / SITESTATUS / REACH* / MEASURE* / FeatureURL	Retained GIS/source-inventory attributes from the Hawai'i StreamStats gauge inventory.
Unnamed leading CSV column	Some tables were saved with pandas index=True; leading blank-header field is a row index, not a scientific variable.

7. Specific archived output values

Watershed	Target T (yr)	Q at target (ft^3/s)	Target stage (ft)	Cutoff (ft)
Hanalei	1000	48,743.4	22	23
Waiahole	1000	21,747.8	21	22
Waialua	1000	37,096.6	35	36

These values are transcribed from the archived watershed_probability_summary.csv; they are not newly recomputed.

Waialua model-versus-rating stage errors

Archived table: 14 model runs; 14 valid signed errors. Mean bias = -0.350 ft; mean absolute error = 0.700 ft; RMSE = 0.862 ft.

Flow (ft^3/s)	GSSHA stage (ft)	Rating stage (ft)	Error (ft)	AEP
1,815	21.588	21.896	-0.308	0.59044
2,931	23.001	23.709	-0.708	0.39789
3,959	23.967	25.165	-1.198	0.28401
5,036	25.041	26.259	-1.218	0.20478
6,519	26.205	26.843	-0.638	0.13571
8,084	27.197	27.404	-0.207	0.09159
10,033	28.226	28.048	+0.178	0.05846
12,053	29.105	28.696	+0.408	0.03823
14,744	30.101	29.506	+0.595	0.02290
17,975	31.078	30.407	+0.672	0.01316
22,185	32.075	31.480	+0.594	0.00688
29,212	33.042	33.078	-0.036	0.00268
37,931	33.784	34.812	-1.028	0.00090
46,654	34.343	36.350	-2.008	0.00030

8. PeakFQ source-output inventory

Peak_Fq/ has three weighted folders: Hanalei_weighted, Waiahole_weighted and Waialua_weighted. Each contains the same 12 artifact types below; the PeakFQ program produces these externally, and notebook 1 reads the *_qnt.csv and *_nwis_peak.txt files. This project does not show the commands that originally ran PeakFQ.

Filename pattern	Purpose
{site}.png	PeakFQ plot image; exact graph content should be checked in the image.
{site}_ema.png	Expected Moments Algorithm (EMA) plot image.
{site}_inp.png	PeakFQ input/diagnostic plot image.
{site}_emp.csv	Empirical frequency/plotting-position table.
{site}_log.txt	PeakFQ run log.
{site}_lp3.csv	Log-Pearson Type III fit output.
{site}_mgb.csv	Multiple Grubbs–Beck test output.
{site}_nwis_peak.txt	NWIS annual peak discharge records; parsed by read_nwis_peak.
{site}_nwis_site.txt	NWIS site metadata text.
{site}_psf.psf	PeakFQ settings/input file.
{site}_qnt.csv	Quantile estimates by exceedance probability; columns EXC_Prob and Estimate feed notebook 1.
{site}_trd.csv	PeakFQ trend-test output.

Hanalei_weighted

16103000.png; 16103000_ema.png; 16103000_inp.png; Hanalei_weighted_emp.csv; Hanalei_weighted_log.txt; Hanalei_weighted_lp3.csv; Hanalei_weighted_mgb.csv; Hanalei_weighted_nwis_peak.txt; Hanalei_weighted_nwis_site.txt; Hanalei_weighted_psf.psf; Hanalei_weighted_qnt.csv; Hanalei_weighted_trd.csv

Waiahole_weighted

16294100.png; 16294100_ema.png; 16294100_inp.png; Waiahole_weighted_emp.csv; Waiahole_weighted_log.txt; Waiahole_weighted_lp3.csv; Waiahole_weighted_mgb.csv; Waiahole_weighted_nwis_peak.txt; Waiahole_weighted_nwis_site.txt; Waiahole_weighted_psf.psf; Waiahole_weighted_qnt.csv; Waiahole_weighted_trd.csv

Waialua_weighted

16210500.png; 16210500_ema.png; 16210500_inp.png; Waialua_weighted_emp.csv; Waialua_weighted_log.txt; Waialua_weighted_lp3.csv; Waialua_weighted_mgb.csv; Waialua_weighted_nwis_peak.txt; Waialua_weighted_nwis_site.txt; Waialua_weighted_psf.psf; Waialua_weighted_qnt.csv; Waialua_weighted_trd.csv

9. Gauge metadata / geospatial files

PULL_USGS_METADATA/Get_stream_metadata.ipynb generates or edits the gauge inventory from Hawai‘i StreamStats and USGS sources. The archived PULL_USGS_METADATA/Hawaii_StreamStats_All_Gages/ folder includes a CSV, GeoJSON, and shapefile component set for Hawaii_StreamStats_All_Gages; and a CSV, GPKG, and shapefile component set for Hawaii_StreamStats_Gages_Peak_Record.

Shapefile .shp, .shx, .dbf, .prj and .cpg are companion parts of a single dataset; do not treat them as independent maps. GPKG is a separate GIS container. These are metadata/spatial input assets, not rating-curve plots.

10. Figure gallery (saved images from archive)

Each figure below is an actual PNG contained in the supplied ZIP, not a newly regenerated model result.

Figure 1. Curves_datum_differences.png

Comparison of old/new Waialua posterior median curves and estimated datum shift. Source: 2_OBSERVED_USGS-rating-curves.ipynb · cell 13

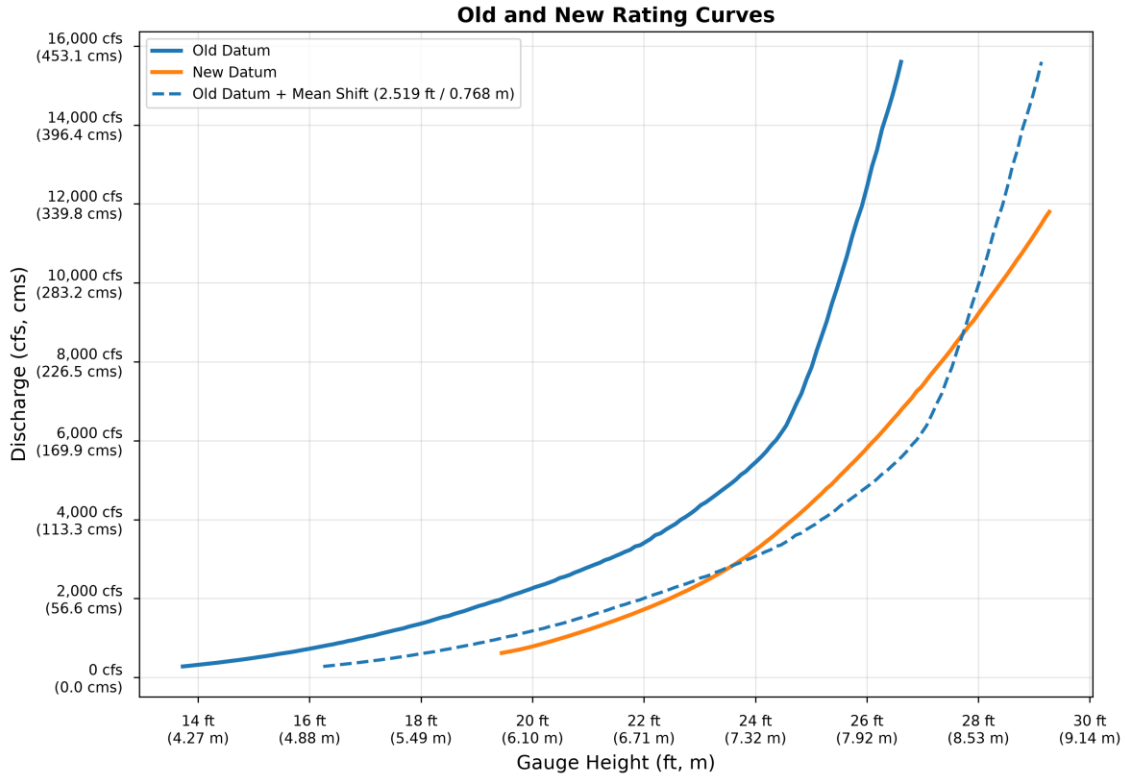


Figure 2. Gauge_datum_differences.png

Stage difference vs discharge for datum adjustment. Source: 2_OBSERVED_USGS-rating-curves.ipynb · cell 14

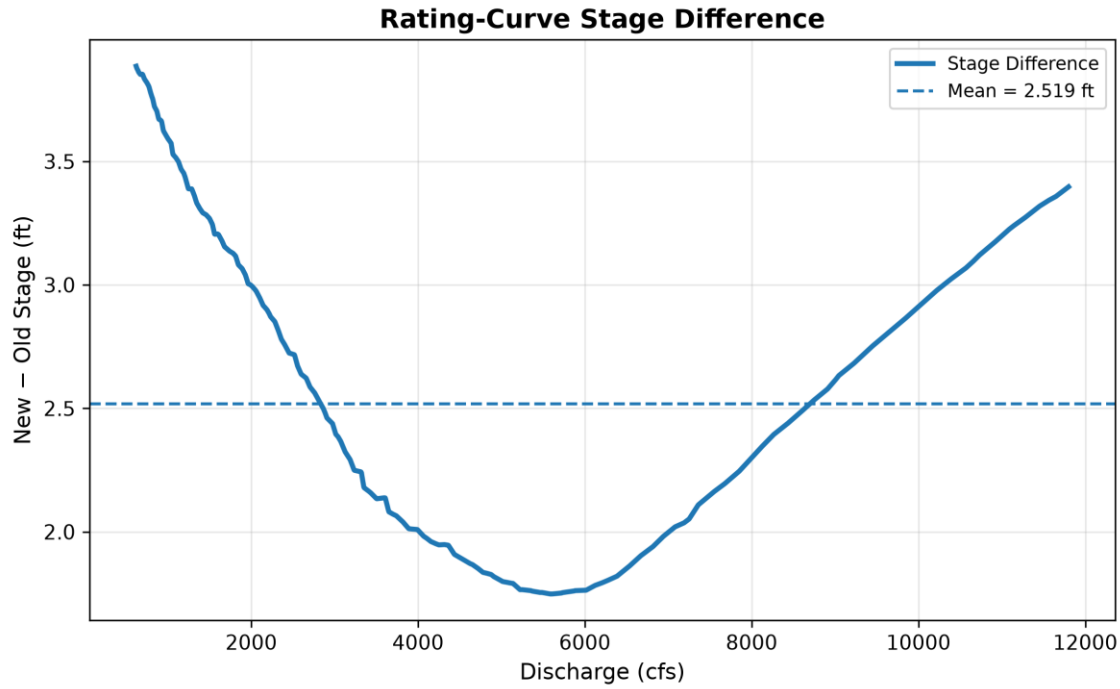


Figure 3. Hanalei_curve.png

Hanalei observed stage–discharge fit, observations and uncertainty. Source: 2_OBSERVED_USGS-rating-curves.ipynb · cell 7

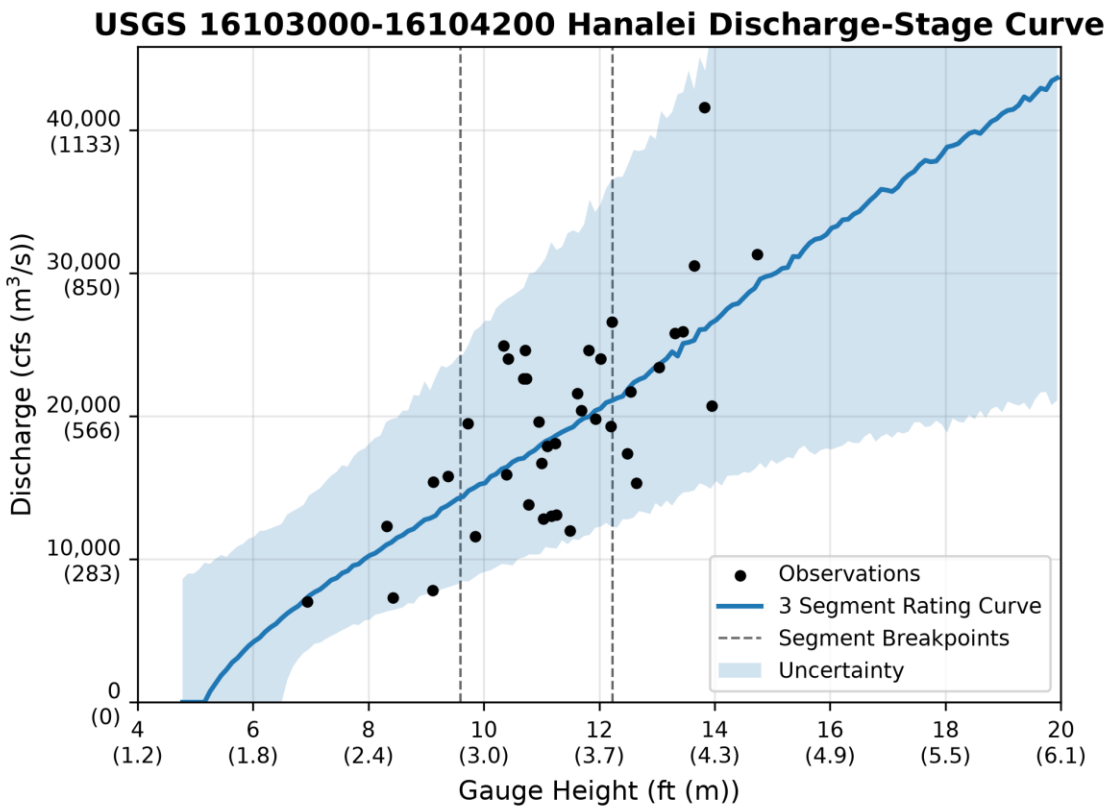


Figure 4. merged_discharge_probability_curves.png

Three-watershed discharge versus annual exceedance probability; PeakFQ points and merged rating values. Source: 3_OBSERVED_probability-stage-discharg.ipynb · cells 3–4

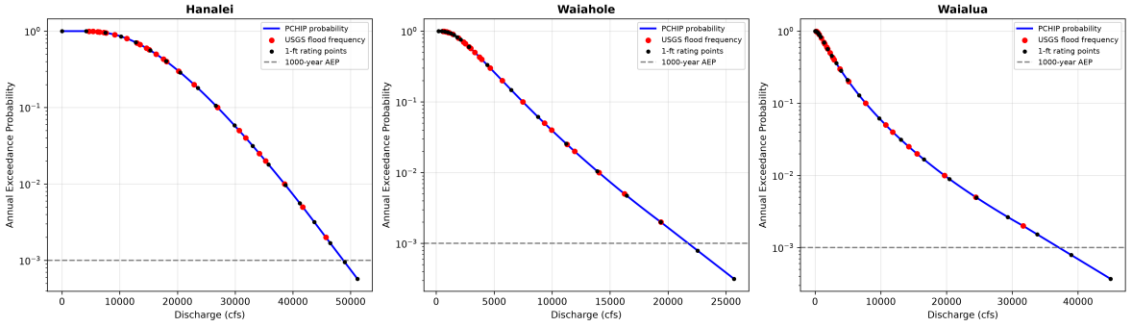


Figure 5. merged_stage_discharge_rating_curves.png

Three-watershed observed rating curves, merged 1-ft stage points and uncertainty. Source: 3_OBSERVED_probability-stage-discharg.ipynb · cells 3–4

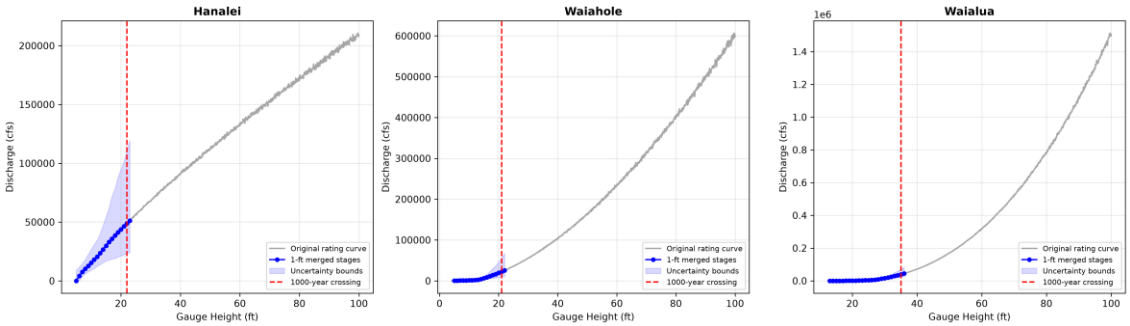


Figure 6. Waiahole_curve.png

Waiahole observed stage–discharge fit. Source: 2_OBSERVED_USGS-rating-curves.ipynb · cell 5

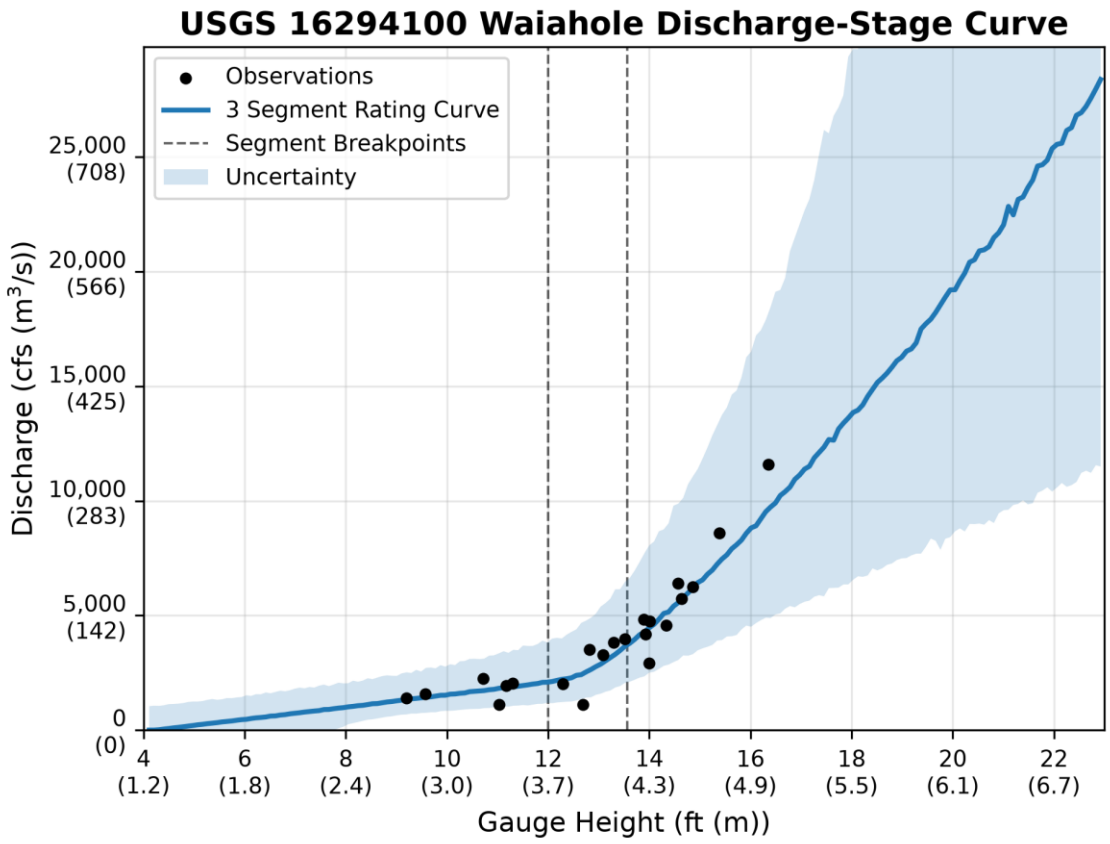


Figure 7. Waialua_Model_Curve.png

Existing archived figure; provenance not established by inspected current source. Source: No direct saving call found in active notebooks

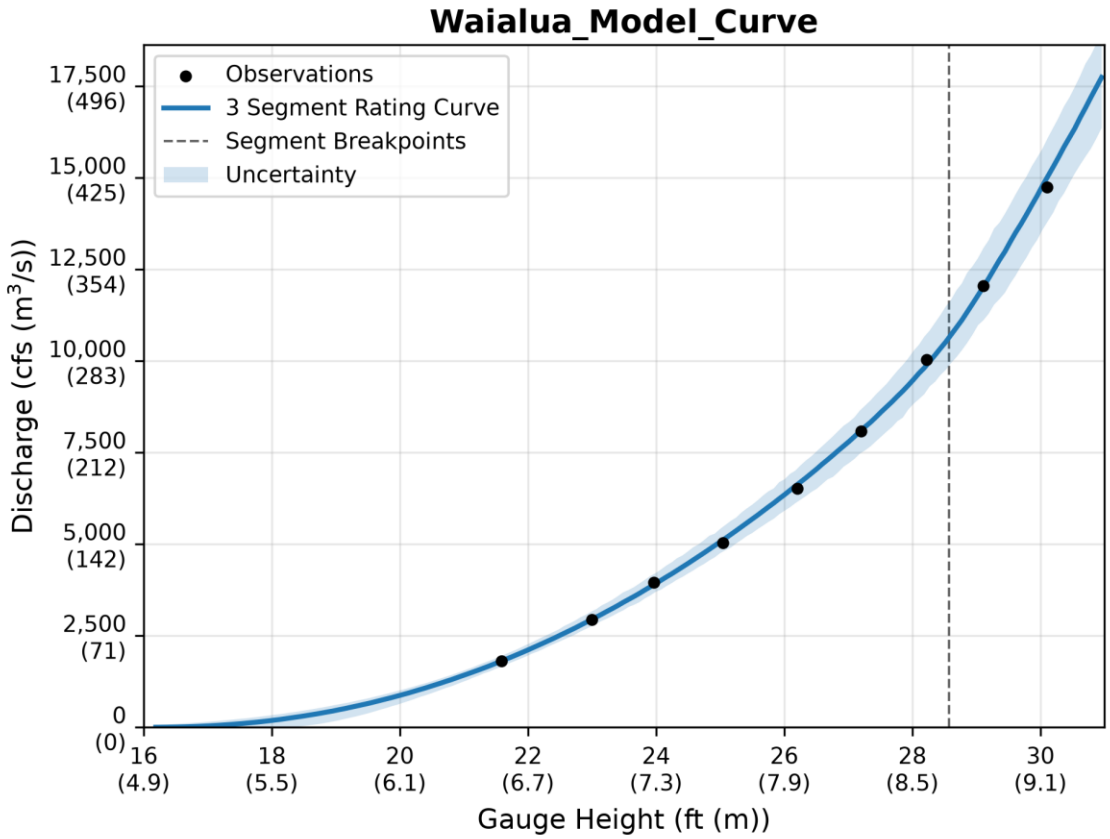


Figure 8. Waialua_newer_datum_curve.png

Waialua newer-datum rating fit. Source: 2_OBSERVED_USGS-rating-curves.ipynb · cell 10

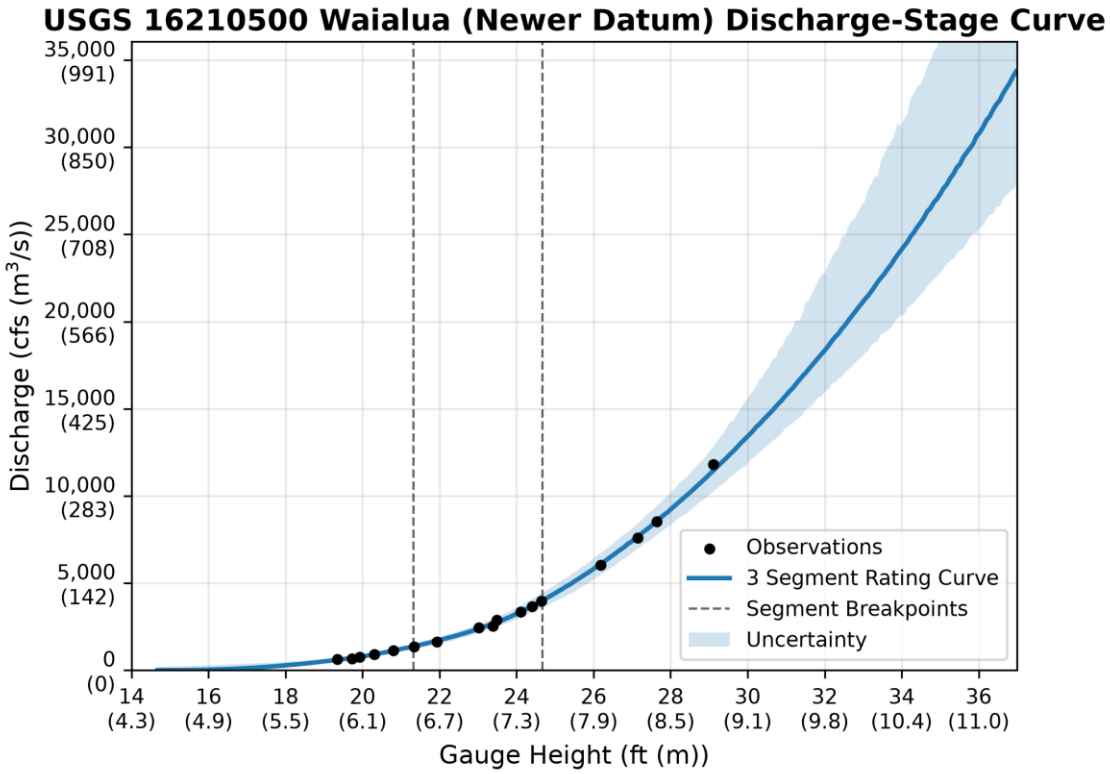


Figure 9. Waialua_older_datum_rating.png

Waialua older-datum rating fit. Source: 2_OBSERVED_USGS-rating-curves.ipynb · cell 11

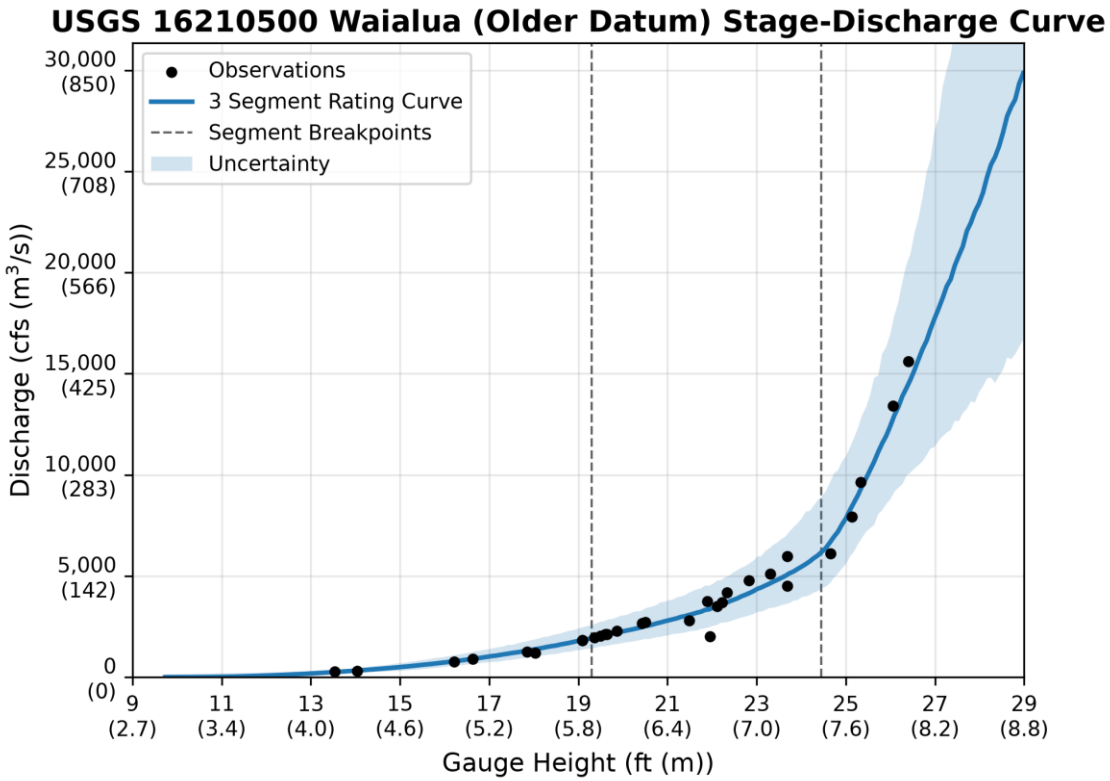


Figure 10. Waialua_rating_curve_adjusted_datums.png

Final Waialua merged datum-adjusted observed rating fit. Source: 2_OBSERVED_USGS-rating-curves.ipynb · cell 18

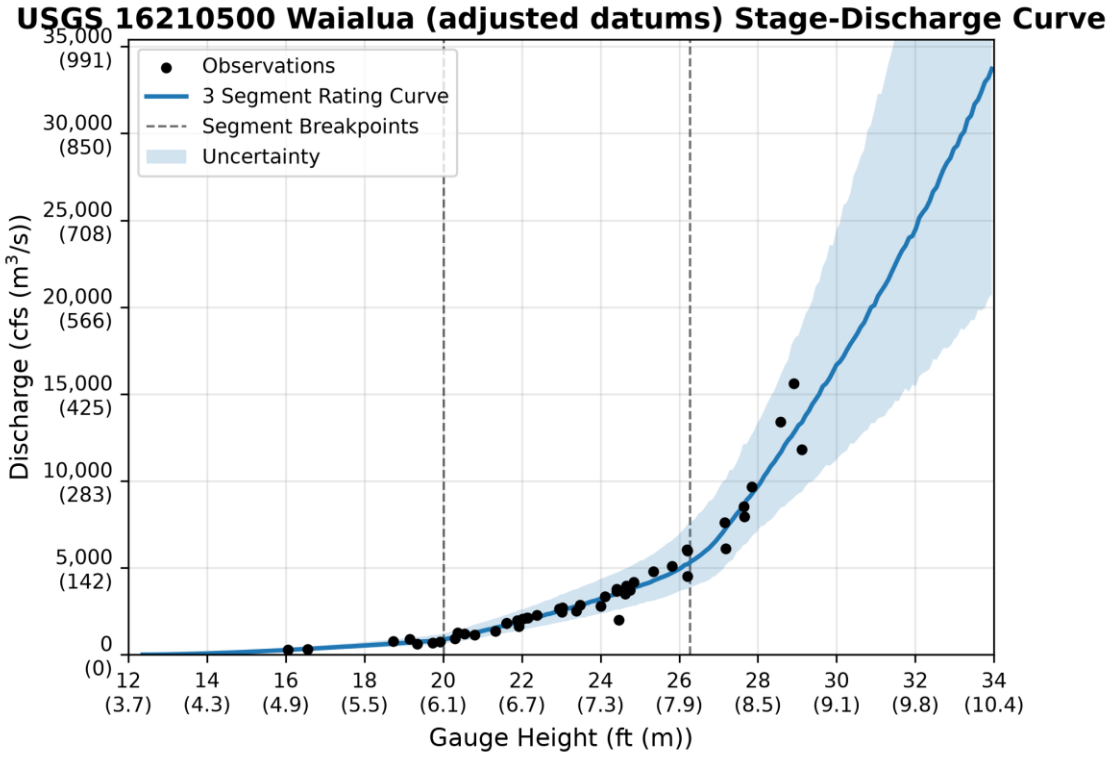
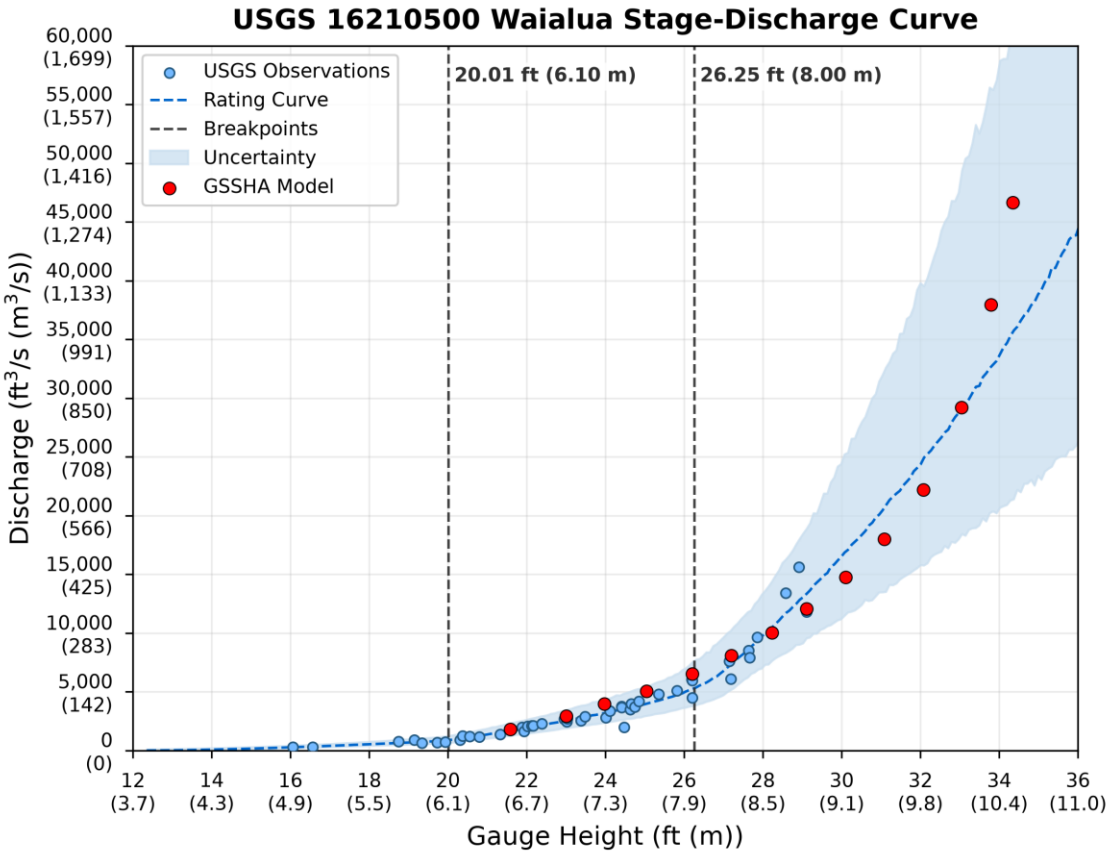


Figure 11. Waialua_rating_curve_comparative_GSSHA.png

GSSHA model points overlaid on observed fitted rating curve and uncertainty. Source: 4_GSSHA-MODEL_compared_w_OBSERVED.ipynb · cell 12 (also 4_READ...)



11. Reproducibility and caveats

• Notebook 4 exists under two filenames with matching content in the archive; treat as parallel copies unless intentionally diverged.

• Compare_MODEL_vs_OBSERVED_rating_probability.py defines get_gssha_rating_curve twice. The later definition (with offset_m and increment) overrides the earlier one when imported.

• Figure Waialua_Model_Curve.png exists but no active notebook save call for that exact filename was found. It may be a legacy output.

• Saved observed rating-curve CSVs contain 1,000 posterior grid rows. The full fitting metadata (segment parameters and posterior draws) are not serialized in those CSVs; the in-memory meta dict is required for plot breakpoints and stage-error reference table.

• The GSSHA stage-error routine uses a 1-ft modeled reference table and inverse PCHIP without extrapolation. Stage errors are differences against the fitted observed curve, not standalone GSSHA predictive uncertainty or confidence intervals.

• Some 95% discharge bounds are generated over a wide stage range (up to 100 ft in one discussed fit); treat out-of-observation-range curves as extrapolative, not observed validation.

• Several output CSVs retain an unnamed pandas index column; step 4 GSSHA FIM CSV is saved without index=False.

• Figure style and saved figures can differ from later notebook states if the code was edited after figures were generated. The archive is a snapshot, not a tested reproducible build.

• PeakFQ quantile columns and some GIS fields are source-specific. Interpret exact PeakFQ ancillary fields using the PeakFQ output definitions; this README does not infer unverified field semantics.

12. File-by-file quick locator

Non-checkpoint files in ZIP: 108. The register above includes every CSV beneath SAVE_DATA/, every saved figures/ PNG, all three Peak_Fq weighted output folders, the principal notebooks, modules, and GIS source families. Excluded from the scientific register: .ipynb_checkpoints/, __pycache__/, and .gitattributes.